```
    return 0;
}
anirudh@LAPTOP-H9V5A2PN:~/OSLAB$ nano 52.c
anirudh@LAPTOP-H9V5A2PN:~/OSLAB$ gcc m52.c -o m52
cc1: fatal error: m52.c: No such file or directory
compilation terminated.
anirudh@LAPTOP-H9V5A2PN:~/OSLAB$ cat m52.c
cat: m52.c: No such file or directory
anirudh@LAPTOP-H9V5A2PN:~/OSLAB$ gcc 52.c -o 52
52.c:1:3: error: invalid preprocessing directive #Makefile
    1 | # Makefile
      |   ^~~~~~~~~
52.c:3:3: error: invalid preprocessing directive #Build
    3 | # Build all shell project modules
      |   ^~~~~
52.c:5:3: error: invalid preprocessing directive #Commands
    5 | # Commands:
      |   ^~~~~~~~~
52.c:6:3: error: invalid preprocessing directive #make
    6 | # make
      |   ^~~~
52.c:7:3: error: invalid preprocessing directive #make
    7 | # make clean
      |   ^~~~
52.c:8:3: error: invalid preprocessing directive #make
    8 | # make test
      |   ^~~~
52.c:9:3: error: invalid preprocessing directive #make
    9 | # make valgrind
      |   ^~~~
52.c:10:3: error: invalid preprocessing directive #make
   10 | # make run-pipeline
      |   ^~~~
52.c:11:3: error: invalid preprocessing directive #make
   11 | # make run-signals
      |   ^~~~
52.c:13:1: warning: data definition has no type or storage class
   13 | CC = gcc
      | ^~
52.c:13:1: warning: type defaults to ‘int’ in declaration of ‘CC’ [-Wimplicit-int]
52.c:13:6: error: ‘gcc’ undeclared here (not in a function)
   13 | CC = gcc
      |      ^~~
52.c:15:1: error: expected ‘,’ or ‘;’ before ‘CFLAGS’
   15 | CFLAGS = -Wall -Wextra -g
      | ^~~~~~~
52.c:36:32: error: stray ‘@’ in program
   36 |         $(CC) $(CFLAGS) $< -o $@
      |                                ^
52.c:39:32: error: stray ‘@’ in program
   39 |         $(CC) $(CFLAGS) $< -o $@
      |                                ^
52.c:42:32: error: stray ‘@’ in program
   42 |         $(CC) $(CFLAGS) $< -o $@
      |                                ^
52.c:45:32: error: stray ‘@’ in program
   45 |         $(CC) $(CFLAGS) $< -o $@
      |                                ^
52.c:48:32: error: stray ‘@’ in program
   48 |         $(CC) $(CFLAGS) $< -o $@
      |                                ^
52.c:51:32: error: stray ‘@’ in program
   51 |         $(CC) $(CFLAGS) $< -o $@
      |                                ^
52.c:54:32: error: stray ‘@’ in program
   54 |         $(CC) $(CFLAGS) $< -o $@
      |                                ^
52.c:57:32: error: stray ‘@’ in program
   57 |         $(CC) $(CFLAGS) $< -o $@
      |                                ^
52.c:60:32: error: stray ‘@’ in program
   60 |         $(CC) $(CFLAGS) $< -o $@
      |                                ^
```

```
anirudh@LAPTOP-H9V5A2PN:~/OSLAB$ cat 52.c
 Makefile

 Build all shell project modules

 Commands:
 make
 make clean
 make test
 make valgrind
 make run-pipeline
 make run-signals

C = gcc

FLAGS = -Wall -Wextra -g

IN = bin

ROGRAMS = \
        $(BIN)/integrated_shell_modules \
        $(BIN)/signal_handler_management \
        $(BIN)/foreground_interrupt_handling \
        $(BIN)/suspend_resume_jobs \
        $(BIN)/process_group_terminal_control \
        $(BIN)/pipeline_stability \
        $(BIN)/pipeline_monitor \
        $(BIN)/memory_cleanup_test \
        $(BIN)/error_scenario_testing

ll: $(BIN) $(PROGRAMS)

(BIN):
        mkdir -p $(BIN)

(BIN)/integrated_shell_modules: src/integrated_shell_modules.c
        $(CC) $(CFLAGS) $< -o $@

(BIN)/signal_handler_management: src/signal_handler_management.c
        $(CC) $(CFLAGS) $< -o $@

(BIN)/foreground_interrupt_handling: src/foreground_interrupt_handling.c
        $(CC) $(CFLAGS) $< -o $@

(BIN)/suspend_resume_jobs: src/suspend_resume_jobs.c
        $(CC) $(CFLAGS) $< -o $@

(BIN)/process_group_terminal_control: src/process_group_terminal_control.c
        $(CC) $(CFLAGS) $< -o $@

(BIN)/pipeline_stability: src/pipeline_stability.c
        $(CC) $(CFLAGS) $< -o $@

(BIN)/pipeline_monitor: src/pipeline_monitor.c
        $(CC) $(CFLAGS) $< -o $@

(BIN)/memory_cleanup_test: src/memory_cleanup_test.c
        $(CC) $(CFLAGS) $< -o $@

(BIN)/error_scenario_testing: src/error_scenario_testing.c
        $(CC) $(CFLAGS) $< -o $@

est: all
        chmod +x tests/test_shell.sh
        chmod +x tests/run_all_tests.sh
        ./tests/test_shell.sh
        ./tests/run_all_tests.sh

algrind: $(BIN)/memory_cleanup_test
        valgrind --leak-check=full --show-leak-kinds=all ./$(BIN)/memory_cleanup_test

un-pipeline: $(BIN)/pipeline_stability
        ./$(BIN)/pipeline_stability

un-monitor: $(BIN)/pipeline_monitor
        ./$(BIN)/pipeline_monitor

un-signals: $(BIN)/signal_handler_management
        ./$(BIN)/signal_handler_management

un-jobs: $(BIN)/suspend_resume_jobs
        ./$(BIN)/suspend_resume_jobs

un-shell: $(BIN)/integrated_shell_modules
        ./$(BIN)/integrated_shell_modules

lean:
```

```
fun-jobs fun-shell
anirudh@LAPTOP-H9V5A2PN:~/OSLAB$ # docs/syscalls.txt

================================================
              SYSTEM CALLS USED
================================================

fork()
Creates a new process.

execvp()
Replaces the current process with another program.

waitpid()
Waits for child process state changes.

pipe()
Creates an inter-process communication channel.

dup2()
Duplicates file descriptors for redirection.

open()
Opens files.

read()
Reads data from a file descriptor.

write()
Writes data to a file descriptor.

close()
Closes file descriptors.

kill()
Sends signals to processes or process groups.

sigaction()
Registers signal handlers.

setpgid()
Creates or assigns a process group.

getpgrp()
Returns the process group ID.

tcsetpgrp()
Transfers terminal control to a process group.

tcgetpgrp()
Gets the current terminal process group.

malloc()
Allocates dynamic memory.

free()
Releases dynamic memory.

getrusage()
Retrieves resource usage information.

clock_gettime()
Measures execution time.

chdir()
Changes the current working directory.

getcwd()
Gets the current working directory.
```